

LARGE LANGUAGE MODELS IN HUMAN-MACHINE INTERACTION

Ömer Emre Mutlu

emre.mutlu@rwth-aachen.de

Abstract

This document serves as a template for the seminar write-up on Large Language Models in human-machine interaction. It will be replaced with the final abstract once the paper is complete.

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1 Introduction

In high-stakes environments such as aerospace, medicine, disaster response, military applications, and similar domains, the effectiveness and speed with which operators interact with the tools available to them are of critical importance. Traditionally, human-machine interaction (HMI) in such domains has been achieved through methods that have been field-tested over many years, such as user interfaces (UIs) utilizing buttons, joysticks, haptic-feedback sensors, and touchscreens. Depending on the application domain, specific norms and regulations govern the permissible interaction mediums for HMI user interfaces, including standards such as IEC 60204-1 for industrial electrical equipment and NASA-STD-3001 for aeronautical systems.[Int16][CWF23]

However, with the introduction of transformer encoder-decoder models and the subsequent popularization of large language models such as Google's BERT, OpenAI's GPT-3, and more recently GPT-5, a wide range of new possibilities has emerged in the field. Moreover, recent advancements in large language models in the areas of speech-to-speech and visual language models have created additional and particularly interesting opportunities.

This paper presents a literature review of these recent advancements in relation to human-machine interaction in the context of safety-critical working environments.

2 Related Work

Very short section; I will be generally discussing how I have made the research paper selection. How I have classified the papers, my methodology, etc. maybe 1–2 paragraphs.

3 State of the Art: LLM-Driven Interaction Technologies

A more interesting section for me as I work at Sagel AI regarding realtime telephony agents that can take agentic actions. I will give a general overview of the general Multimodal LLM Interaction Technologies.

3.1 Text-Based Interfaces, Planning, and Tool Use

This is plain text interfaces similar to ChatGPT or LLaMA or DeepSeek.

3.2 Speech-to-Text, LLM and Text-to-Speech

This is how speech-to-speech interfaces were being done a few years ago by translating input speech into text, feeding it into an LLM, and then turning the resulting output into speech again. It is a solid approach, however quite slow. Easier to integrate with RAG though.

3.3 Speech-to-Speech LLMs

This is the state of the art regarding speech-to-speech man-machine interactions ever since the launch of OpenAI Realtime models. It works basically by taking input speech in divided chunks

and feeding them into the model as they come. The model starts outputting directly after the first few chunks of audio come in, giving it a near realtime feel during conversation. Very fast, however cognitively a lot dumber than the top-of-the-line models such as GPT-4o or GPT-5. Quite hard to integrate with RAG but not impossible. There are a few papers I saw regarding this but never done it myself; currently working on this at work.

3.4 Vision-Language(-Action) Models

I have learned of this sort of model family not so long ago, and while looking for papers for this seminar I came across these again. Basically this model family can draw context from visual streams such as video feeds and take action. I need to read more on this to give a better explanation but certainly very interesting in context of this seminar topic.

4 Domain Case Studies _____

In this section I will be discussing select papers that are relevant to this seminar topic. Exactly which papers are to be determined. Potential research direction is given below.

4.1 Aerospace and Remote Science

Aerospace, Deep Sea wayages etc.

Psychological and Cognitive Impacts

I am sure in long isolated voyages the use of a somewhat human-feeling interface is a lot better from a psychological aspect rather than a GUI. I have found a paper regarding this. It is certainly something I would like to involve in this paper.

4.2 Emergency and Environmental Response

4.3 Industrial Operations

5 Comparative Analysis and Metrics _____

I will be comparing the methods I have found in the Domain Case Studies section.

6 Safety, Trust, and Deployment Constraints _____

Safety is a huge topic regarding the use of AI in any system that may endanger lives. And hallucinations are a huge issue. I will be looking into more papers in this direction. At work we had huge problems with our customer because the realtime speech models hallucinate on the very extreme end. Several times they were making things up regarding our customers' data. Ours is a CRM system; in the worst case our customers would have to deal with angry clients, and such a thing can result in deaths in aerospace etc.

specieal working environments, local environments what kind of LLM technologies be used in special environments no internet, no electricity etc etc.

7 Conclusion

This section will be my conclusion.

Bibliography

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